

Analytical methods for camera-trap species interactions — paper-by-paper synthesis

Prepared for Fundación Mar Adentro (Bosque Pehuén). Single site, ~26–27 stations, 6 focal species (3 native carnivores, 3 invasive), modest per-cell detections, effort not yet compiled as an exact trap-night/detection-history matrix.

Each block: full reference (as heading), abstract in brief, the problem the authors identify, the method, methodological cautions, limitations, and a verdict on fitness for FMA data with the reasoning. Source for all synthesis below is the attached PDFs themselves; where I state a numeric threshold it comes directly from the paper's text.

Note on one file: the "Estimation of population density by spatially explicit..." PDF is **Efford (2011), *Methods in Ecology and Evolution*** — the *area-search* SECR paper, **not** the classic camera-trap SECR (Efford 2004 / Royle et al. 2009). This matters for feasibility (see that block).

MacKenzie, D.I., Nichols, J.D., Lachman, G.B., Droege, S., Royle, J.A. & Langtimm, C.A. (2002). Estimating site occupancy rates when detection probabilities are less than one. *Ecology* 83(8): 2248–2255.

Abstract in brief. Non-detection $\neq$ absence unless detection probability = 1. The paper proposes a likelihood-based model to estimate the proportion of sites occupied (ψ) while jointly estimating per-visit detection probability (p) from repeated presence/absence visits. Covariates on both ψ and p are supported; missing visits are allowed. Simulation shows estimates are good and generally unbiased when $p > \sim 0.3$. Illustrated with two frog species at 32 wetlands.

Problem identified. Naive occupancy (proportion of sites where the species was seen) is negatively biased because species go undetected when present. This biases any downstream inference about distribution or habitat association.

Method. Build a detection history (a string of 1/0 per site over T repeat occasions), then fit a two-parameter hierarchical model: ψ (occupancy) and p (detection), each optionally a logit-linear function of covariates. Maximum likelihood; standard errors via nonparametric bootstrap (they warn the asymptotic SE is a lower bound and too small at small n). This is the foundational single-season occupancy model.

Methodological cautions. Requires the **closure assumption** (occupancy state fixed during the season). Requires **repeat occasions** per site — you must discretize your deployment into sampling occasions (e.g. 3–7 day windows) to build the detection history. Estimates degrade badly when p is low (< 0.3), which is a real risk for puma and güiña at modest effort. No false-positive detections assumed.

Limitations. Single-season, single-species; says nothing about interactions between species or about time-of-day. It is a *prerequisite* layer, not an interaction method.

Verdict for FMA — feasible and foundational, but you must build the effort/detection-history matrix first. This is the gateway to almost everything in bucket B and to Rota/Doser. Your stated blocker (effort only approximated; no true per-station start/end dates) is exactly what has to be resolved to use it. With ~26 stations and modest detections you can fit $\psi(.)p(.)$ or $\psi(\text{altitude})p(\text{effort})$ for the better-detected species (culpeo, hare, boar, dog); the two low-detection natives (puma, güiña) will have wide CIs and possibly unstable p. **Highest-priority enabling step, not an endpoint.**

MacKenzie, D.I., Nichols, J.D., Hines, J.E., Knutson, M.G. & Franklin, A.B. (2003). Estimating site occupancy, colonization, and local extinction when a species is detected imperfectly. *Ecology* 84(8): 2200–2207.

Abstract in brief. Extends the 2002 model to **multiple seasons**, estimating initial occupancy plus **colonization** (γ) and **local extinction** (ϵ) probabilities between seasons, without assuming process stationarity, using a Pollock robust-design sampling structure. Illustrated with Northern Spotted Owls and tiger salamanders; fit in program PRESENCE.

Problem identified. The 2002 closure assumption fails across years. Inferring colonization/extinction from a single snapshot ("incidence functions") confounds process with pattern — many processes yield the same occupancy pattern.

Method. Primary periods (seasons) each contain secondary occasions (closed within-season). Model estimates ψ_1 , then season-to-season γ and ϵ , with p estimated throughout. This is the **dynamic (multi-season) occupancy model**.

Methodological cautions. Needs several seasons to estimate transition rates with any precision — dynamic parameters are data-hungry. Robust-design structure (repeat occasions nested in seasons) must be respected. Same $p > \sim 0.3$ concern.

Limitations. Still single-species. γ/ϵ precision requires more primary periods than you currently have.

Verdict for FMA — conceptually relevant, not yet feasible. You have 3 campaigns (Otoño 25, Prim-ver 25–26, Otoño 26 in pipeline), with gaps (no Invierno) and seasonal-not-annual structure. Three irregular seasons is thin for stable γ/ϵ , and colonization/extinction of a wide-ranging puma across a 26-station single grid is hard to interpret. **Park this until you have more seasons and a clean effort matrix.** Its real value now is conceptual: it frames "intraspecific temporal shifts across campaigns" as a dynamic-occupancy question you can revisit later.

Rota, C.T., Ferreira, M.A.R., Kays, R.W., Forrester, T.D., Kalies, E.L., McShea, W.J., Parsons, A.W. & Millsbaugh, J.J. (2016). A multispecies occupancy model for two or more interacting species. *Methods in Ecology and Evolution* 7: 1164–1173.

Abstract in brief. Generalizes single-species occupancy to **two or more interacting species** by treating the latent occupancy state as a multivariate Bernoulli variable.

Crucially, it does **not require the asymmetric-interaction assumption** of earlier two-species models (i.e. it doesn't force one species to be "dominant"/conditional on the other). Co-occurrence probability can itself be modeled as a function of environmental covariates. Demonstrated on bobcat, coyote, grey fox, red fox across 6 US states, with human-disturbance covariates; some pairs' co-occurrence shifted along the disturbance gradient (e.g. coyote and grey fox independent at low disturbance, co-occurring at high).

Problem identified. Prior co-occurrence occupancy models were practically limited to two species and often assumed one species' occupancy is conditional on the other's — biologically awkward for symmetric competition. Ignoring interactions biases habitat-association inference.

Method. Symmetric multi-species occupancy with imperfect detection; occupancy states modeled via multinomial logit / multinomial probit (Gibbs sampler given for the probit). Interaction parameters capture whether species co-occur more/less than expected, and covariates can move that. Implemented in `unmarked` (`occuMulti`) and later in `spOccupancy` (next paper).

Methodological cautions. The state-space dimension grows as $2^{(\text{number of species})}$ — for your 6 species that is 64 latent states. The authors are explicit that finding a parsimonious model gets hard fast because of the many pairwise and higher-order interaction terms; they recommend variable-selection techniques. **This is fundamentally a spatial co-occurrence model — it does NOT use time-of-day.** It answers "do they occupy the same stations more/less than expected," not "do they avoid each other in time."

Limitations. Data-hungry per added species and per interaction covariate; sparse cells (few detections) destabilize higher-order interaction terms. Needs the same detection-history + effort scaffolding as MacKenzie 2002.

Verdict for FMA — high value, feasible in a restricted form. This is the single best fit for your core spatial question ("do invasives displace natives in space, and how do the 3 natives partition among themselves") because it handles imperfect detection and symmetric interactions and lets altitude enter as a covariate. **But do not fit all 6 species at once with your sample sizes** — the 2^6 state space plus sparse puma/guñía cells will not support it. Fit **targeted pairs or small triples**: e.g. guñía-dog, culpeo-hare, and the 3 native carnivores as a triple, each with altitude (and season) as covariates. Detection-history matrix is a prerequisite. **One of your 2-3 highest-value additions.**

Doser, J.W., Finley, A.O., Kéry, M. & Zipkin, E.F. (2022). `spOccupancy`: An R package for single-species, multi-species, and integrated spatial occupancy models. *Methods in Ecology and Evolution* 13: 1670–1678.

Abstract in brief. Software paper. `spOccupancy` fits single- and multi-species occupancy models, including **spatially explicit** versions that account for spatial autocorrelation, in a Bayesian framework using Pólya-Gamma data augmentation for fast inference. Supports data integration across datasets (joint likelihood), spatial random effects via Nearest-

Neighbour Gaussian Processes (scales to 1,000s–100,000s of sites), plus model validation (posterior predictive checks), comparison (WAIC, k-fold CV), and prediction.

Problem identified. The models above (MacKenzie, Rota, spatial extensions) existed but lacked unified, computationally efficient, well-documented software — especially for spatial autocorrelation and large data.

Method. Bayesian MCMC with efficient data augmentation; NNGP for spatial structure. Implements the **Rota multi-species model** (`spMsPG0cc`, `msPG0cc`) and single-species spatial/non-spatial variants. User-friendly formula interface, simulation functions, and cross-validation built in.

Methodological cautions. Bayesian workflow means priors, convergence diagnostics (R-hat, trace plots), and posterior predictive checks are on you — more analyst responsibility than a canned MLE fit. The spatial (NNGP) machinery is designed for large site counts; with **26 stations spatial autocorrelation is barely estimable** and the spatial layer will mostly add uncertainty rather than signal.

Limitations. It's an implementation, not a new estimator — inherits every data requirement of the underlying occupancy models. Its headline scalability advantage is irrelevant at your grid size.

Verdict for FMA — use it as the engine, skip the spatial bells and whistles. This is very likely the package you'd actually run Rota-style multi-species co-occupancy in (cleaner than `unmarked`) for the Bayesian multi-species case, with built-in WAIC/CV). Use the **non-spatial** multi-species functions; do **not** invoke the NNGP spatial layer with 26 stations. Practical tooling recommendation rather than a distinct method. Pairs directly with the Rota block.

Rowcliffe, J.M., Kays, R., Kranstauber, B., Carbone, C. & Jansen, P.A. (2014). Quantifying levels of animal activity using camera trap data. *Methods in Ecology and Evolution* 5: 1170–1179.

Abstract in brief. Introduces a method to estimate **activity level** — the *proportion of the 24 h day an animal is active* — from the distribution of camera-trap detection times, by fitting a flexible circular (von Mises / weighted kernel) distribution and computing the fraction of time active from it. Validated by simulation and against independent estimates for small-to-medium mammals. Implemented in the `activity` R package (`fitact`).

Problem identified. Activity level (energetics, foraging effort, risk exposure) is hard to measure non-invasively in the field. Overlap curves describe *shape* but not the scalar *proportion active*.

Method. Assumes detection frequency at a given time is proportional to the number of animals active then; fits a circular kernel; derives activity level. Bootstrap CIs. Gives a **defensible scalar with a CI** rather than only a curve — this is precisely the "effect size / null" gap you flagged with Dhat4.

Methodological cautions. Key assumption: all individuals are active at the peak of the daily cycle. The authors explicitly warn this is less likely met in large predators (→ puma is the risky case) and in high-latitude winters / small mammals in cold environments (they excluded torpid small-mammal-in-cold cases as outliers). On sample size: CV was ~21% at n=42 and dropped to ~10% by n≈100. So ~100 detections for a stable estimate; below ~40 it's shaky.

Limitations. Single-species metric; comparing activity levels between species is an indirect route to temporal segregation, not a direct overlap test. Assumption violation in large predators biases the estimate upward.

Verdict for FMA — feasible and complementary for your better-sampled species; treat puma with caution. This directly extends your existing von Mises overlap work by adding a scalar activity level with CIs — useful for "does species X compress or shift its active time across campaigns" (your intraspecific seasonal-shift question). Realistic for culpeo, hare, boar, dog where you can approach ~100 detections; marginal for güiña and puma on current n, and the peak-activity assumption is explicitly weakest for puma. Use it as a complement to overlap, per-species where n allows. A solid, low-cost addition.

Frey, S., Fisher, J.T., Burton, A.C. & Volpe, J.P. (2017). Investigating animal activity patterns and temporal niche partitioning using camera-trap data: challenges and opportunities. *Remote Sensing in Ecology and Conservation* 3(3):123–132.

Abstract in brief. A review/framework paper (not a new estimator). Surveys the analytical toolkit for activity patterns and temporal niche partitioning from timestamped camera data, and argues the field has described temporal overlap well but rarely tests how biotic + abiotic drivers (climate, landscape change, community composition, disturbance) shape temporal partitioning. Advocates coordinated distributed experiments and, critically, integration of spatial and temporal analyses.

Problem identified. Most studies stop at describing overlap; few link temporal-niche shifts to mechanisms, and spatial vs temporal analyses are usually run separately rather than jointly.

Method. Conceptual synthesis: it frames terminology (activity level vs activity pattern vs overlap), classifies approaches, and lays out a research agenda. No package of its own — it points to overlap/kernel tools and to the need for spatiotemporal integration.

Methodological cautions. As a review it gives direction, not a validated procedure. Its "coordinated distributed experiments" prescription assumes multi-site designs you don't have.

Limitations. Non-quantitative; you cannot "run" Frey. Value is in framing and in justifying why you should push toward joint spatiotemporal methods (bucket C).

Verdict for FMA — read it as the framing/justification document, not a method.

Excellent for your introduction/discussion and for motivating the move beyond pairwise overlap toward joint spatiotemporal analysis and driver-based tests (altitude, native-vs-invasive guild, season). It essentially *names the gap* your project is trying to fill. **Cite it to frame; don't expect an analysis from it.**

Karanth, K.U., Srivathsa, A., Vasudev, D., Puri, M., Parameshwaran, R. & Kumar, N.S. (2017). Spatio-temporal interactions facilitate large carnivore sympatry across a resource gradient. *Proceedings of the Royal Society B* 284: 20161860.

Abstract in brief. Empirical study of dhole, leopard, tiger across four Indian reserves spanning a prey/carnivore density gradient. Tests **temporal, spatial, and combined spatio-temporal** segregation for each pair. Finds the three axes don't mirror each other — species use them as *alternative* mechanisms — and that overlaps are higher at low prey density. Combined spatio-temporal overlap was minimal (possibly by chance); fine-scale avoidance appeared at one high-density reserve.

Problem identified. Temporal overlap alone, or spatial overlap alone, can miss how species actually avoid each other; you need to combine axes and compare across a resource gradient.

Method (a genuine methodological menu you can borrow):

- **Temporal:** von Mises fits per species; Watson U^2 to test if activity patterns differ; bootstrap CIs on proportions of exclusive/overlapping active time.
- **Spatial:** MacKenzie **two-species occupancy** with the **Species Interaction Factor (SIF** $= \psi_{AB} / (\psi_A \cdot \psi_B)$); SIF < 1 = segregation, > 1 = aggregation, = 1 = independence; AIC comparison of SIF-estimated vs SIF-fixed-at-1 models. (Note: they had to simplify — the two-species detection-interaction model **did not converge**, so they set detection independent. A direct warning about data hunger at your scale.)
- **Spatio-temporal:** a **site × hour detection matrix**, plus a **time-to-encounter / minimum-time-to-co-predator** statistic tested against a null generated by **randomizing encounter times across camera locations (1000 permutations, MRPP)**. Observed > expected time-to-encounter = segregation; < expected = aggregation. This null is *conditional on observed space-use and activity*, which is exactly the defensible-null property you wanted.

Methodological cautions. Their two-species *detection*-interaction model failed to converge even with 99–180 stations per reserve and 30 days — a strong signal that full interaction detection models are out of reach at 26 stations. Time-to-encounter needs enough co-detections at shared stations to be meaningful. The cross-reserve gradient design is central to their inference and you have no landscape gradient.

Limitations. Their strongest conclusions lean on the density gradient across four reserves — a multi-site contrast you can't replicate at one site. Combined spatio-temporal overlap was so low they attribute it partly to chance, underscoring how data-hungry the joint signal

is.

Verdict for FMA — the closest template to your goal; borrow the time-to-encounter permutation approach, scale down the rest. This is the paper that operationalizes bucket C. The **time-to-encounter / randomization test** is directly transferable to a single site and gives you a defensible null and effect direction (avoidance vs attraction) for native-invasive pairs at shared stations — your biggest current gap. The **SIF two-species occupancy** is usable pairwise (with detection independence, as they were forced to). What you *cannot* reproduce is the resource-gradient cross-reserve comparison. **Adopt the spatio-temporal permutation method; it's one of your 2-3 highest-value additions.** Feasibility hinges on having enough co-detections at shared stations — check that per pair before committing.

Niedballa, J., Wilting, A., Sollmann, R., Hofer, H. & Courtiol, A. (2019). Assessing analytical methods for detecting spatiotemporal interactions between species from camera trapping data. *Remote Sensing in Ecology and Conservation* 5(3): 272-285.

Abstract in brief. A simulation benchmark of methods for detecting two types of interaction: **spatiotemporal avoidance** (a species avoids a site *after* another has been there) and **temporal segregation** (shifts in daily activity). For avoidance they compare linear models, Mann-Whitney U, a permutation test, and a random-records test on the time intervals between species' records; for temporal segregation they use a permutation test. They report statistical power, validity of p-values, and **minimum sample sizes** across activity patterns and interaction strengths. Framework tied to **camtrapR** (corresponding author is its maintainer).

Problem identified. Many methods exist for "do species interact in space-time," but there was no comparative assessment of which is powerful and valid, or of the sample sizes needed — so people pick methods blindly.

Method / headline results (directly useful numbers):

- **Linear models** on (log) time-intervals-between-records were the **most powerful and flexible** approach for spatiotemporal avoidance, and can correct for other sources of variation (e.g. station as a fixed/random effect).
- Detecting spatiotemporal **avoidance** reliably needed **~50 records per species** (power strongly driven by interaction strength).
- Detecting strong **temporal segregation** needed **far fewer — ~10-20 records —** but power depended heavily on the underlying activity pattern.
- **All tests were valid** (uniform p-values under the null) down to a **minimum of ~10 records per species**.
- Imperfect detection is expected to **reduce power for avoidance** tests but should **not bias the temporal-segregation** test given sufficient records.

Methodological cautions. The ~50-records-per-species bar for avoidance is the binding constraint for your low-detection natives. U-tests are a lower-power fallback when linear-

model assumptions fail. The simulations are two-species; extending to imperfect detection and multi-species is future work they flag.

Limitations. Two-species, simulation-based; real-world messiness (uneven effort, clustered detections) not fully captured. But the sample-size guidance is exactly the strict feasibility input you asked for.

Verdict for FMA — read this as your feasibility ruler, and adopt the linear-model interval approach. This paper effectively tells you *which* of your bucket-C ideas will work at your n: **temporal segregation tests are feasible even at your low counts (~10–20 records)**, while **spatiotemporal avoidance needs ~50 records per species**, which several native-invasive pairs won't reach for the natives. The **linear-model-on-time-intervals** method (implementable around camtrapR outputs) is the recommended, most-powerful tool and lets you fold in station and effort as covariates. **Use its thresholds to decide, per pair, whether avoidance is even testable.** One of your highest-value additions precisely because it's honest about when to stop. Feasibility: temporal segregation — yes; spatiotemporal avoidance — only for the better-detected pairs.

Efford, M.G. (2011). Estimation of population density by spatially explicit capture-recapture analysis of data from area searches. *Methods in Ecology and Evolution* 2: 654–662.

Abstract in brief. Presents likelihood-based **spatially explicit capture-recapture (SECR)** for **area-search** data — repeated searches of polygonal plots (or transects) where individuals (or identifiable sign) are located. Shows the method is at least as robust as an equivalent Bayesian analysis, with negligible bias and near-nominal CI coverage on a lizard dataset. Recommends search plots be **at least as large as the target species' home range**. Implemented in the `(secr)` R package.

Problem identified. SECR methods had focused on *passive* detectors (traps, cameras at fixed points). Some species/data suit *active area searches* (polygon searches, sign belonging to identified individuals) — needing a different detection model.

Method. Likelihood SECR for binary or count detections over irregular polygons/transects, estimating density from the spatial pattern of re-detections of identified individuals. **Requires individual identification.**

Methodological cautions. This is the **area-search variant**, not the point-detector camera-trap SECR you'd normally use for a camera grid. Individual ID is mandatory. Plot/detector geometry must relate sensibly to home-range size. For a *camera grid* you would instead use point-detector SECR (Efford 2004; Royle et al. 2009) — also in `(secr)/(oscr)` — which this paper is adjacent to but not the canonical citation for.

Limitations. Density estimation needs individually identifiable animals with enough spatial recaptures. Of your 6 species only **güiña** is plausibly individually identifiable (spotted); puma and the rest are not reliably ID-able. Even for güiña, 26 stations at a single site with modest detections likely yields too few spatial recaptures for a stable density

estimate.

Verdict for FMA — largely out of scope as written; density via SECR is a stretch even for güiña. Individual ID rules out 5 of 6 species immediately. For güiña, point-detector SECR (not this area-search paper) *could* in principle give density, but you'd need enough recaptures of identified individuals across stations — unlikely at current effort. This paper is the *area-search* flavour, which doesn't match a fixed camera grid at all. **Treat SECR as aspirational for a future güiña-focused, higher-effort sub-study; not a near-term option, and if pursued, cite the point-detector SECR papers, not this one.** Also note the REM (Rowcliffe et al. 2008) density-without-ID route you mentioned in your brief is *not* among these nine PDFs.

Cross-cutting: ranked shortlist for a single Andean site (~26 stations)

Ranked by value-added given your constraints. "Prereq" = needs the exact effort / detection-history matrix built first.

1. **Rota multi-species co-occurrence occupancy** — Rota et al. 2016, run in `spOccupancy` (non-spatial) or `unmarked::occuMulti`. *Why*: directly answers native-vs-invasive spatial exclusion *and* native-native partitioning with imperfect detection and altitude as covariate. Fit **pairs/small triples**, not all 6. **Prereq**.
2. **Time-to-encounter / spatiotemporal permutation test** — Karanth et al. 2017 approach. *Why*: fills your biggest gap (joint space-time avoidance) with a defensible randomization null at a single site. Feasible where pairs have enough shared-station co-detections.
3. **Linear-model on inter-record time intervals + temporal-segregation permutation test** — Niedballa et al. 2019 (around `camtrapR`). *Why*: most powerful avoidance test *and* tells you honestly when n is too low; temporal segregation is testable even at ~10–20 records. Use its thresholds as a go/no-go per pair.
4. **Activity-level estimation with CIs** — Rowcliffe et al. 2014, `activity` package. *Why*: upgrades your von Mises overlap work to a scalar with CIs for intraspecific seasonal shifts. Good for culpeo/hare/boar/dog; cautious for güiña/puma.
5. **Single-season occupancy (ψ, p)** — MacKenzie et al. 2002, `unmarked`/`spOccupancy`. *Why*: the enabling layer for #1 and a defensible replacement for naïve occupancy with altitude covariate. **Prereq — and building this matrix unblocks the top of the list.**

Not yet / out of scope: dynamic multi-season occupancy (MacKenzie 2003 — too few seasons); SECR density (Efford 2011 area-search — needs individual ID; only güiña qualifies and recaptures will be sparse). **Frey 2017** is framing, not a method — use it to motivate the joint spatiotemporal push.

The single most unblocking action: compile true per-station deployment start/end dates into an effort/detection-history matrix. It converts items 1 and 5 from "blocked" to "runnable" and sharpens 2–4.

